

IMA205 Challenge

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1 Introduction

White blood cells (WBCs) are key components of the immune system, and their morphological properties — such as size, shape, nuclear structure, and cytoplasmic texture — provide critical information for diagnosing a variety of diseases, including leukaemia and infections. Traditionally, WBC classification is performed manually by haematologists — microscopic examination of stained blood smears, a process that is time-consuming and subject to inter-observer variability.

Recent advances in machine learning have enabled the development of automated systems for medical image analysis, offering improved scalability and reproducibility. However, WBC classification remains challenging due to high intra-class variability, subtle inter-class differences, and potential class imbalance.

The objective of this work is to classify microscopy images into 13 distinct WBC categories, using a training dataset of 28,901 images. Performance is evaluated using the macro-averaged F1 score, which ensures balanced consideration across all classes.

This report follows a structured, hypothesis-driven approach, where preprocessing techniques, training strategies, and model architectures are explored with clear motivations grounded in the characteristics of the data. Each design choice is informed by both prior knowledge and empirical observations.

2 Exploratory Data Analysis

The training dataset comprises 28,901 images of white blood cells (WBCs) distributed across 13 classes. Several of these classes correspond to the same underlying cell lineage, and in some cases represent different stages of cellular maturation.

For instance, within the neutrophil lineage, the dataset distinguishes between band-form neutrophils (BNE) and segmented neutrophils (SNE). More broadly, the myeloid lineage includes progressive maturation stages such as promyelocytes (PMY), myelocytes (MY), and metamyelocytes (MMY). In the lymphoid lineage, the dataset differentiates between typical lymphocytes (LY), variant (atypical) lymphocytes (VLY), and prolymphocytes (PLY), reflecting morphological variations rather than a strict maturation sequence.

This class structure introduces hierarchical relationships and high inter-class similarity, which makes the classification task inherently challenging.

I divided the original training dataset into training and validation sets using a weighted, stratified random split, with 80% of the data allocated to training and the remaining 20% to validation.

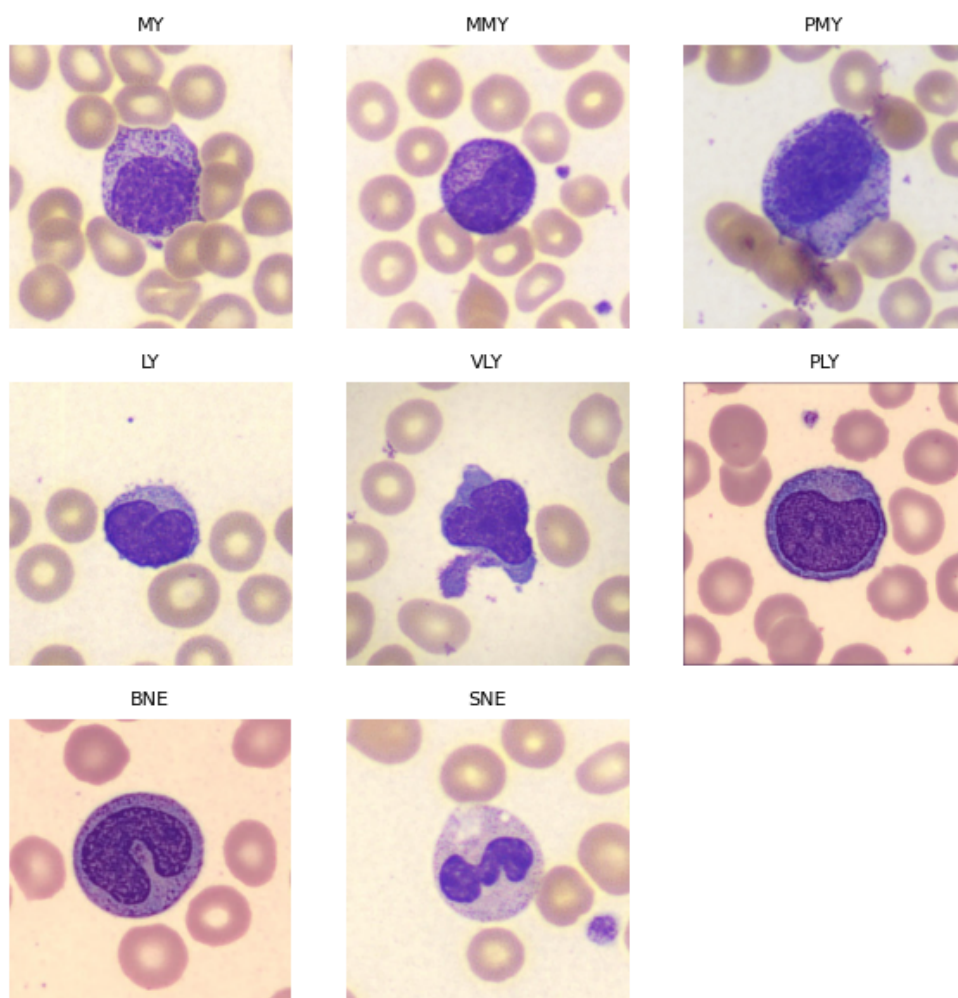


Figure 1: Examples of each class of myelocytes, lymphocytes and neutrophils

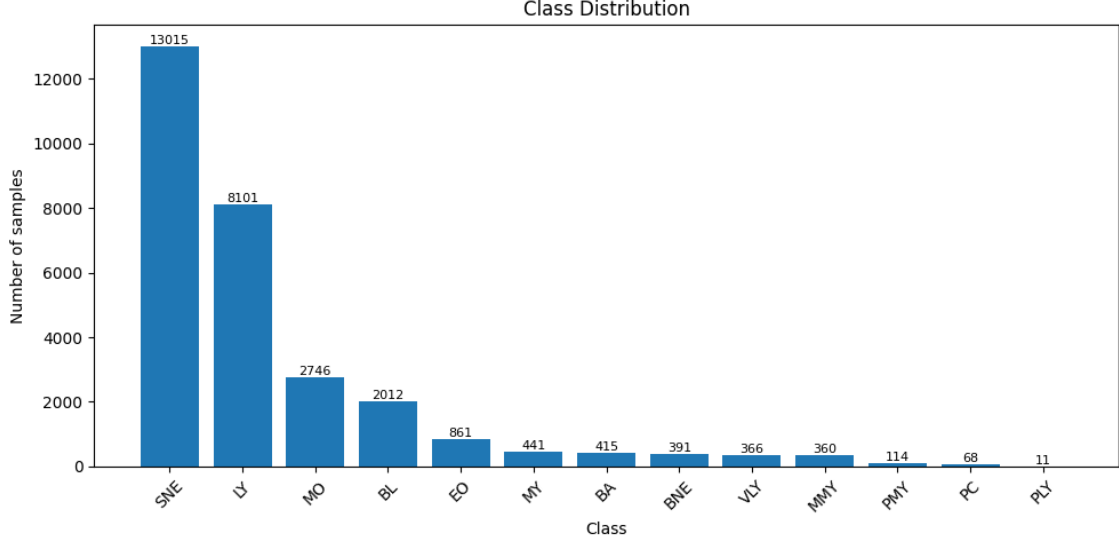


Figure 2: Class distribution for training set

The dataset is highly imbalanced as can be seen in figure 2. Furthermore, for each type of cell that is divided in subtypes one always dominates except for myelocytes.

All this suggests that dealing with class imbalance is one of the main problems here and that possibly using specialists to distinguish between clusters could improve performance.

Furthermore, visual inspection of the dataset revealed that a subset of images exhibited significant noise, which could potentially hinder model performance. To systematically assess this, I quantified image noise using a simple median-filter residual metric, where noise is estimated as the average absolute deviation between the original image and its median-filtered version.

This choice is motivated by the nature of the noise observed, which is predominantly impulsive (salt-and-pepper-like). Median filtering is well known to be robust to such noise, as it preserves structural content while suppressing isolated pixel-level outliers. As a result, the residual $|I - \tilde{I}|$ effectively captures high-frequency, non-structural perturbations, making it a simple yet appropriate proxy for image corruption.

The distribution of noise scores (Figure 3) exhibits a strong right-skew, with a large concentration of samples near zero and a long tail extending toward higher values. This indicates that the majority of images are relatively clean, while a smaller subset exhibits progressively higher levels of degradation. Notably, the distribution suggests natural separation regimes rather than a continuous uniform spread.

Based on this observation, I discretized the noise into three categories: clean, noisy, and very noisy, using thresholds at $t_1 = 20$ and $t_2 = 40$. The first threshold separates the dense low-noise region from moderately corrupted samples, while the second threshold isolates the extreme tail of high-noise images.

Furthermore, noise is relatively uniform across classes. Slightly higher levels are observed in myeloid precursor classes (e.g., MY, MMY), while classes such as PC exhibit lower noise; however, these differences are modest. Importantly, classes with very small sample sizes (e.g., PLY, PC) show more variability, but these estimates are statistically unreliable due to limited data.

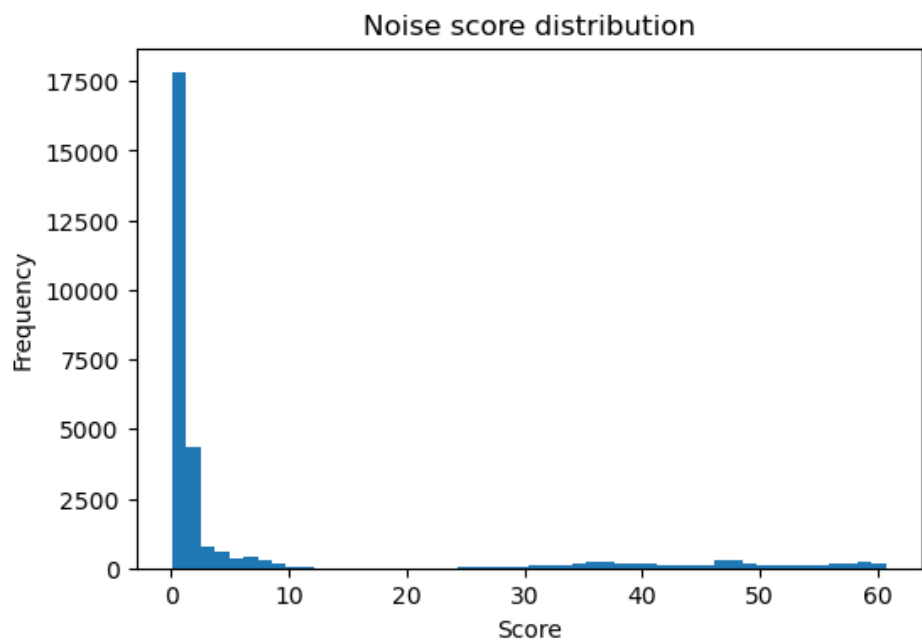


Figure 3: Distribution of median-filter residuals

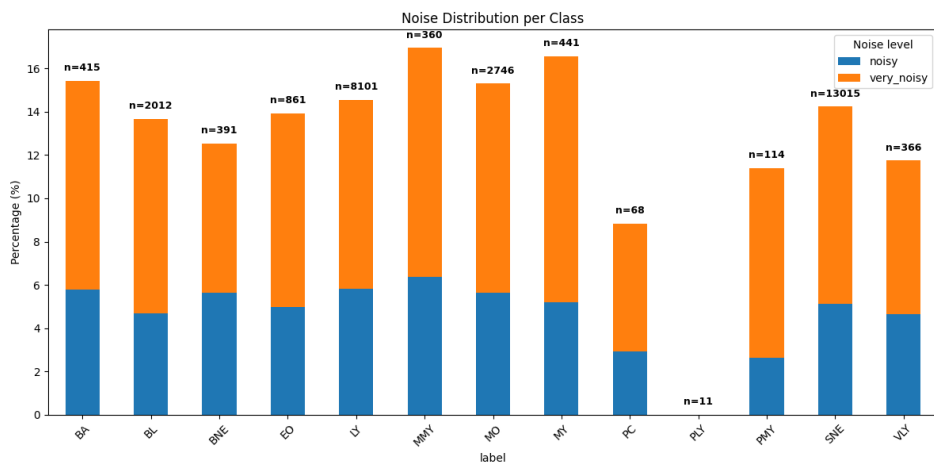


Figure 4: Percentage of noisy images per class

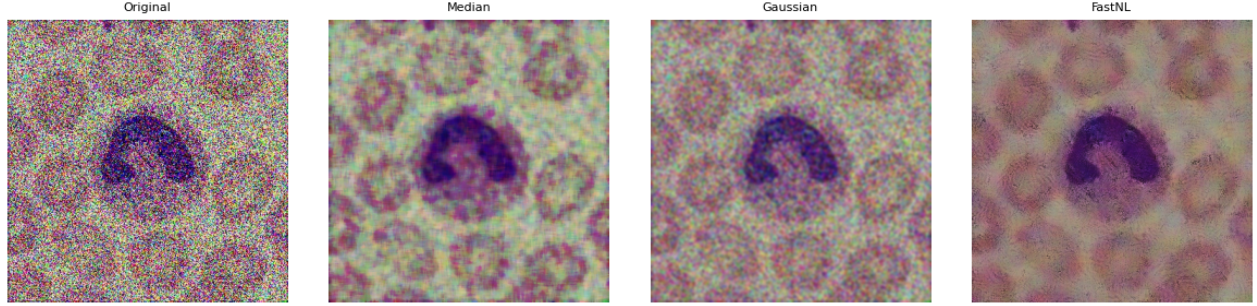


Figure 5: Comparison of denoising methods (manually tuned)

3 Preprocessing

Here I present all the preprocessing methods used throughout the project

3.1 Denoising

The denoising method used, `fastNlMeansDenoisingColored`, is based on the non-local means (NLM) algorithm. Unlike local filters (e.g., Gaussian or median), NLM computes each pixel as a weighted average of pixels within a larger search region, where weights depend on patch similarity:

$$\hat{I}(i) = \sum_{j \in \mathcal{N}(i)} w(i, j) I(j), \quad w(i, j) \propto \exp \left(-\frac{\|P(i) - P(j)\|^2}{h^2} \right).$$

This allows the method to suppress noise while preserving edges and fine structures by favoring contributions from similar patches.

The `fastNlMeansDenoisingColored` variant extends this approach to color images, using separate parameters (h , h_{color}) for luminance and chrominance components. The parameter h controls the smoothing strength: larger values increase noise removal but risk over-smoothing.

For images classified as *noisy*, moderate values ($h = 20$, $h_{color} = 20$) were used to balance noise reduction and detail preservation. For *very noisy* images, stronger filtering ($h = 40$, $h_{color} = 40$) was applied to suppress severe artifacts, accepting a slight loss of detail. The template and search window sizes were fixed to (7, 21), providing a standard trade-off between structural fidelity and computational cost.

Overall, this adaptive parameterization enables effective noise suppression while preserving the subtle morphological features that are critical for fine-grained WBC classification.

3.2 Cropping

Motivated by the need to resize all inputs to a fixed resolution for the deep learning models, I hypothesized that cropping around the cell would preserve more fine-grained morphological details by allocating a larger proportion of the input pixels to the region of interest. I expected this to be particularly beneficial for distinguishing between visually similar cell types, where subtle differences in nucleus shape, chromatin texture, or cytoplasmic structure are critical.

Center cropping. A fixed 224×224 crop centered on the image was extracted. This simple heuristic leverages the dataset bias that cells are typically located near the center, providing a robust baseline without requiring explicit localization.



Figure 6: Example of a cropped cell

HSV-based cropping. The image was converted to the HSV color space and thresholded to isolate purple/blue regions corresponding to the nucleus. The largest contour was selected and used to define a bounding box, with additional padding to retain contextual information. This approach encodes prior knowledge about color distributions but relies on fixed thresholds, making it sensitive to staining variability.

LAB-based cropping. A more adaptive strategy was implemented in the LAB color space. The *A* channel was normalized and thresholded using a percentile-based criterion, followed by morphological filtering and contour extraction. A randomized margin was applied to the detected region, introducing variability in crop size and acting as a form of implicit data augmentation.

However, the experimental results suggest that this effect is limited in practice. We obtained basically the same performance, the models appear to retain sufficient discriminative information even after resizing the full image, likely due to their ability to learn robust, scale-invariant features. Additionally, the loss of contextual information introduced by aggressive cropping may partially offset the potential gains in local detail. On the other hand, the errors were not exactly the same for models trained with noisy vs denoised dataset, which was useful for the building of ensembles.

4 Initial Pipeline

I drew inspiration from [1], which shows that convolutional neural networks can be effectively leveraged as *feature extractors* rather than purely end-to-end classifiers for white blood cell (WBC) classification. In particular, the study demonstrates that combining deep representations with classical machine learning models can outperform fully end-to-end approaches.

Motivated by this, I adopt a **decoupled pipeline** in which deep models are used exclusively for feature extraction, followed by a separate classification stage. Specifically, I extract features from multiple pretrained architectures, namely ResNet-50 [3] and ConvNeXt-Base [2]. These models are trained under different preprocessing configurations (e.g., denoising, cropping), and their feature representations are concatenated to form a unified embedding space.

This design serves two purposes. First, it leverages the strong representational power of modern deep networks while avoiding the need to train a large end-to-end classifier on a relatively small and imbalanced dataset. Second, combining features from different architectures introduces complementary inductive biases: ResNet captures hierarchical patterns through residual learning, while ConvNeXt incorporates modern design choices that improve feature quality and robustness. As a result, the concatenated feature space is both rich and diverse.

To further refine the representation, I perform **feature selection** on the concatenated embeddings. Features are ranked using both mutual information and Ridge-based importance scores, and different subsets are evaluated. This step is critical due to the high dimensionality and redundancy

of the combined feature space. Selecting the most informative features improves both computational efficiency and predictive performance, particularly for models sensitive to collinearity such as QDA.

On top of these extracted features, I train classical machine learning models, namely Quadratic Discriminant Analysis (QDA), linear Support Vector Machines (SVM), and LightGBM. This choice is particularly suitable for the problem setting. First, the dataset exhibits significant class imbalance and limited samples in several categories, making end-to-end training prone to overfitting. Second, classical models offer more controlled decision boundaries: QDA explicitly models class-specific covariance structures, SVM focuses on margin maximization in high-dimensional spaces, and LightGBM captures non-linear interactions through gradient boosting. Third, by decoupling representation learning from classification, the number of trainable parameters in the final stage is drastically reduced, improving stability and generalization.

4.1 Training Feature Extractors

4.1.1 Data augmentation.

To improve generalization and mitigate overfitting, a set of lightweight geometric and photometric augmentations was applied during training. These transformations were designed to reflect plausible variations in microscopy images while preserving the underlying cellular morphology.

Geometric transformations. Random horizontal and vertical flips, as well as rotations by multiples of 90° , were applied. These operations are label-preserving in this setting, as the orientation of white blood cells in microscopy images is arbitrary and carries no diagnostic meaning. Incorporating these transformations encourages the model to learn rotation- and reflection-invariant features, reducing sensitivity to acquisition conditions.

Photometric transformations. A mild color jitter was applied by randomly adjusting brightness and contrast. This simulates variations in staining intensity, illumination, and imaging conditions that naturally occur across samples. The perturbations were intentionally kept small to avoid distorting diagnostically relevant features such as chromatin texture or cytoplasmic color.

Overall, the augmentation strategy aims to increase data diversity while remaining consistent with the physical and biological properties of the data. This acts as a form of regularization and is especially important when using a weighted sampler so that the model does not memorize the minority samples.

4.1.2 Weighting and Upsampling

Class-weighted loss. Using class weights in the cross-entropy loss consistently yielded the best and most stable performance. This approach rebalances the contribution of each class directly at the loss level, increasing the penalty for misclassifying minority classes while preserving the original data distribution. As a result, the model is encouraged to learn discriminative features for rare classes without significantly distorting the input distribution.

Weighted sampling. In contrast, weighted random sampling was slightly less effective. Although it increases the frequency of minority class samples during training, it does so by oversampling a very limited number of images. This can lead to overfitting, especially for extremely small classes, and introduces higher variance in the gradient updates. Additionally, altering the sampling distribution may negatively impact the learning of majority classes, leading to a less balanced overall representation.

Focal loss. Focal loss was also evaluated but performed poorly in this setting. This is likely due to insufficient hyperparameter tuning, as focal loss introduces additional parameters (e.g., the focusing parameter γ) that significantly affect its behavior. Without careful calibration, focal loss

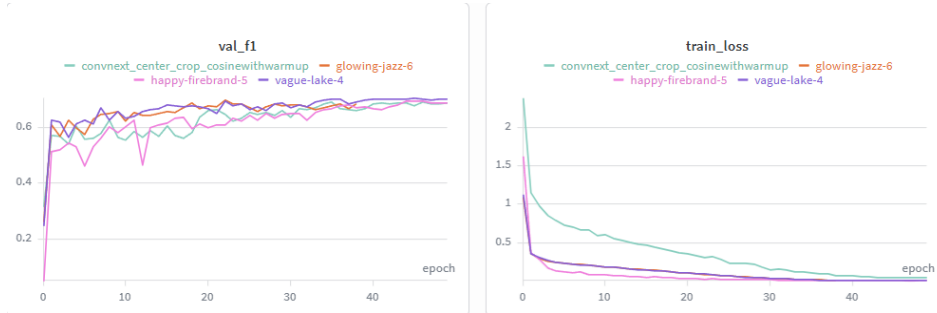


Figure 7: Training of ConvNext under different conditions (pre-processing, weighting and sampling, etc). We observe that overall performance is not very significantly impacted

can overemphasize hard or noisy samples, which is particularly problematic in this dataset where some classes are both rare and visually ambiguous. As a result, it led to unstable training and inferior performance compared to standard class-weighted cross-entropy.

Summary. Overall, class-weighted loss provided the best trade-off between stability and performance, effectively addressing class imbalance without introducing the drawbacks associated with oversampling or poorly tuned loss reweighting.

4.1.3 Schedulers

I found learning rate to be a critical component for achieving stable and high-performing models. Due to the complexity of the task and the variability in the dataset, using a fixed learning rate led to suboptimal convergence, either causing premature stagnation or unstable training dynamics.

To address this, I explored different scheduling strategies and observed that adaptive learning rate decay significantly improved both convergence speed and generalization. In particular, cosine annealing yielded the best overall performance, as it enables smooth and non-monotonic learning rate decay, allowing the model to escape shallow minima and refine solutions in later stages of training. Additionally, an exponential decay scheduler with a decay factor of 0.95 also performed strongly, providing a consistent and controlled reduction of the learning rate that stabilizes training.

4.1.4 Multi-stage fine-tuning

I experimented with a two-stage fine-tuning strategy using ResNet-50, where I first trained only the final classification layer with the backbone frozen, and then fine-tuned the entire network. This approach is commonly motivated by transfer learning best practices: it aims to preserve pretrained features while gradually adapting the model to the target task, which can be beneficial when data is limited or noisy. Given the fine-grained nature of WBC classification, I hypothesized that stabilizing training in this way would help the model generalize better.

However, in practice, two-stage tuning underperformed compared to naive end-to-end training. A likely explanation is the strong domain shift between natural images and microscopy data: freezing the backbone early prevented the model from adapting its features to the specific morphological patterns of white blood cells. As a result, the classifier was trained on suboptimal representations, and later fine-tuning could not fully recover from this misalignment. In contrast, naive tuning allowed simultaneous adaptation of both features and classifier from the start, which proved more effective for this highly specialized and fine-grained task.

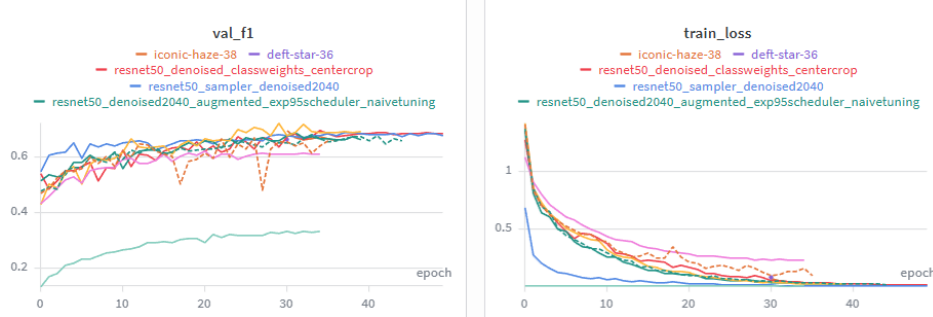


Figure 8: Training of ResNet under different conditions. The one that stands out is focal loss with a bad performance. Some methods are more stable.

4.2 Feature selection and hyperparameter tuning

4.2.1 Feature selection

Feature selection is particularly important in this setting for two reasons. First, high-dimensional feature spaces can lead to overfitting, especially when combined with relatively small class sizes and imbalanced data. Second, many of the extracted features are likely to be correlated or non-informative, as they originate from different models observing similar visual patterns. Reducing the feature space can therefore improve both generalization and computational efficiency. To this end, I employed two complementary ranking methods:

Mutual Information This method measures the statistical dependency between each feature and the class labels, capturing potentially non-linear relationships. Features with higher mutual information are more informative for predicting the target variable.

Ridge-based ranking A Ridge classifier was trained on the full feature set, and features were ranked according to the magnitude of their coefficients. This provides a supervised, model-based importance measure that reflects how strongly each feature contributes to a linear decision boundary, while accounting for correlations through ℓ_2 regularization.

From these rankings, three selection strategies were considered: using the top- k features according to MI, the top- k according to Ridge, and the intersection of both sets. The latter aims to retain features that are consistently identified as important by both a statistical criterion (MI) and a model-based criterion (Ridge), thereby improving robustness and reducing noise.

4.2.2 Hyperparameter tuning

Tuning was conducted on the concatenated deep feature representations after standardization, using a mid-range feature subset ($k = 1000$ selected via mutual information). This setting provides a good trade-off between dimensionality and information content, making it a reliable proxy for general performance.

For QDA, the regularization parameter was varied over $\{0.0, 0.1, 0.2, 0.5\}$ to cover under-, moderate-, and over-regularized regimes, with the best performance obtained at `reg_param` = 0.5. For Linear SVM, the regularization parameter $C \in \{0.1, 1.0, 5.0\}$ was explored to capture different margin-complexity trade-offs, with the best configuration corresponding to $C = 1.0$.

For LightGBM, a focused grid search was performed around standard and computationally efficient configurations:

`num_leaves` $\in \{31, 63\}$, `learning_rate` $\in \{0.01, 0.05, 0.1\}$, `n_estimators` $\in \{100, 200, 300\}$.

The best-performing configuration on the validation set was:

`num_leaves` = 31, `learning_rate` = 0.05, `n_estimators` = 200,

which was subsequently fixed for all experiments.

4.3 Results on Deep Feature Representations

Table 1 reports the performance of different classifiers trained on concatenated features extracted from **ConvNeXt-Base** and **ResNet-50**.

4.3.1 Numerical instability in QDA.

When using more than 200 features, some dimensions of the feature space become linearly dependent, leading to rank-deficient covariance matrices.

QDA relies on inverting Σ_k to compute class-conditional densities. When Σ_k is singular or ill-conditioned, this inversion becomes unstable, resulting in unreliable estimates of the decision boundary. Although the regularization parameter (`reg_param`) partially mitigates this issue by shrinking the covariance matrix toward a diagonal form, it does not fully resolve the instability in high-dimensional, collinear settings.

This behavior is reflected in the experimental results. QDA performs competitively at low dimensionality (e.g., Intersection with $k = 200$), where covariance estimation is more stable. However, as the number of features increases, performance degrades significantly, consistent with the increasing prevalence of rank-deficient covariance matrices. In contrast, discriminative models such as Linear SVM and LightGBM are less sensitive to feature collinearity and high dimensionality, which explains their superior and more stable performance across configurations.

Table 1: Macro F1-score for different feature selection strategies and models.

Selection	k	LightGBM	Linear SVM	QDA
Intersection	200	0.6922	0.6999	0.7014
Intersection	500	0.6943	0.6972	0.6321
Intersection	1000	0.6950	0.6916	0.6285
Intersection	1500	0.6980	0.6884	0.6008
MI	200	0.6795	0.6939	0.6703
MI	500	0.6872	0.6956	0.6488
MI	1000	0.7034	0.6981	0.6082
MI	1500	0.6951	0.7013	0.5132
Ridge	200	0.6921	0.6927	0.6725
Ridge	500	0.6986	0.6945	0.6352
Ridge	1000	0.6919	0.6858	0.5853
Ridge	1500	0.6970	0.6984	0.5491

4.3.2 Effect of feature selection.

Feature selection plays a key role in extracting the most useful information from the concatenated representations. For LightGBM, performance improves as k increases up to 1000, after which it saturates or slightly degrades, suggesting that additional features introduce redundancy or noise.

The intersection strategy performs particularly well at low k , indicating that features consistently ranked important by both MI and Ridge form a compact and robust subset. However, at larger k , this strategy becomes overly restrictive compared to MI.

4.3.3 Model comparison.

LightGBM achieves the best overall performance, benefiting from its ability to model non-linear interactions. **Linear SVM** performs consistently well and remains close to LightGBM, suggesting that the feature space is largely linearly separable. In contrast, **QDA** performs significantly worse and degrades with increasing dimensionality. The frequent occurrence of non-full-rank covariance matrices indicates strong feature collinearity, violating QDA assumptions and leading to numerical instability.

4.3.4 Interpretation.

Overall, these results suggest that the combination of ConvNeXt and ResNet features already captures most of the discriminative information required for the task. The role of feature selection and downstream classifiers is therefore mainly to refine decision boundaries and mitigate redundancy, rather than to extract fundamentally new information. Therefore, I decided to change strategy.

5 Specialist Models for Difficult Clusters

To address confusions between morphologically similar classes, I explored training specialized models on subsets of classes corresponding to particularly challenging clusters. The motivation was that restricting the label space would allow models to focus on fine-grained distinctions and learn more discriminative representations within these regions.

These specialists were trained using the same feature extraction framework based on ResNet-50 and ConvNeXt-Base, leveraging their complementary strengths in capturing hierarchical and fine-grained visual patterns.

Training required careful optimization. In particular, I observed that reducing the learning rate was necessary to obtain stable convergence. This suggests that the task is sensitive to optimization dynamics, likely due to the reduced dataset size and the increased difficulty of fine-grained discrimination within each cluster.

Performance varied significantly across clusters. For the neutrophil and myeloid groups, the specialists achieved macro F1-scores of approximately 0.78 and 0.70, respectively, indicating that the approach can be effective when morphological differences remain sufficiently pronounced. However, the lymphoid cluster proved substantially more challenging, with performance around 0.60 and high variability across runs.

This behavior suggests that lymphoid subclasses are intrinsically harder to distinguish, likely due to subtle and overlapping morphological characteristics. Additionally, the smaller effective dataset size for these classes may exacerbate instability during training.

6 Swin Transformer Backbone

In addition to convolutional architectures (ResNet-50 and ConvNeXt-Base), I incorporated a **Swin Transformer (Swin-B)** [4] as a third feature extractor. The motivation for this choice was to explore whether a transformer-based architecture, with its ability to model long-range dependencies and global context, could better capture subtle morphological patterns in white blood cell images.

Unlike CNNs, which rely on local receptive fields and hierarchical feature aggregation, Swin Transformers operate using self-attention within shifted windows, enabling both local and non-local interactions. This is particularly relevant in this setting, where discriminative features may depend not only on fine-grained textures (e.g., chromatin patterns), but also on broader spatial arrangements (e.g., nucleus-to-cytoplasm ratio or overall cell structure).

Empirically, the Swin-B model achieved slightly better performance than the convolutional backbones, particularly when combined with **LAB-based cropping**. This can be explained by the interaction between the preprocessing and the model architecture. LAB cropping emphasizes the nucleus region—where most diagnostically relevant information is concentrated—by leveraging color-space properties that highlight hematoxylin-stained structures. As a result, the input becomes more focused and semantically aligned.

This localized and structure-aware input appears to benefit the Swin Transformer more than CNN-based models. Since self-attention mechanisms are sensitive to the distribution of information across the image, removing irrelevant background and centering the salient region allows the model to allocate its capacity more effectively. In contrast, CNNs are generally more robust to spatial redundancy due to their inductive biases (e.g., translation invariance and locality), which may explain why cropping had only marginal impact for them.

Overall, the results suggest that transformer-based architectures can better exploit carefully preprocessed inputs, particularly in fine-grained classification tasks where subtle spatial relationships are important. However, the performance gain remains modest, reinforcing the conclusion that representation quality is already high and that improvements are incremental.

7 Ensembles

7.1 Confusion Matrix Analysis and Motivation for Ensembling

The confusion matrices of the best models from each architecture family (ResNet-50, ConvNeXt, and Swin Transformer) reveal a consistent high-level structure, but with important differences at a finer scale. While overall performance is similar across models, their error distributions are not identical.

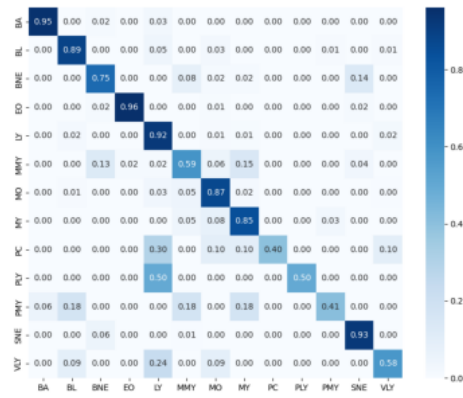
All models achieve near-perfect performance on morphologically distinct classes such as BA, EO, LY, and SNE. In contrast, most errors are concentrated within biologically related groups. In particular, strong confusion is consistently observed between band-form and segmented neutrophils, as well as across myeloid maturation stages (MMY, MY, PMY), reflecting the subtle and continuous nature of these morphological transitions.

However, despite this shared structure, the models differ in how these confusions are expressed. For instance, some models are more prone to confusing specific myeloid stages, while others better preserve distinctions within neutrophil subtypes. Similarly, although lymphoid subclasses (LY, VLY, PLY) are challenging for all models, the degree to which minority classes collapse into LY varies noticeably across architectures. These differences indicate that the models are not making identical mistakes, even when their aggregate metrics are similar. This observation suggests that classification errors are not solely determined by architectural limitations, but rather by intrinsic

ConvNeXt



Swin Transformer



ResNet-50



Figure 9: Confusion Matrices of best model per architecture

ambiguity in the data combined with model-specific inductive biases. Different architectures and preprocessing strategies (e.g., cropping, denoising, or color-space transformations) lead to different feature representations and decision boundaries, which in turn produce complementary error patterns.

Furthermore, this is not only true for the best models of different architectures, but within the same architecture, for models trained differently (denoising vs no denoising, class weights vs weighted sampler, different crops etc) is also true in some cases.

Rather than viewing these differences as inconsistencies, they can be interpreted as a source of diversity. In particular, models that perform similarly at the aggregate level may still capture different aspects of the data distribution and fail on different subsets of examples. This complementarity is precisely the condition under which ensembling is most effective.

Motivated by this, I chose to aggregate multiple models instead of selecting a single best one. By combining models trained with different architectures and preprocessing pipelines, the ensemble can leverage their complementary strengths and reduce individual weaknesses. This is especially beneficial for fine-grained and ambiguous classes, where no single model consistently dominates.

7.2 Methodology

As a first step, I implemented a weighted ensemble, where predicted class probabilities are combined through a weighted average. The weights are optimized via random search over the simplex using Dirichlet sampling, directly maximizing macro F1 on the validation set.

Dirichlet sampling provides an efficient way to explore the space of convex combinations in a continuous and unbiased manner. Unlike grid search, which becomes intractable as the number of models increases, or independent random sampling, which may violate normalization constraints, the Dirichlet distribution enables scalable and well-structured exploration of weight configurations.

Finally, this approach avoids introducing additional optimization complexity or overfitting risks associated with gradient-based tuning, while still allowing the ensemble to adaptively emphasize stronger models and downweight weaker ones. In practice, this simple stochastic strategy proved sufficient to identify high-performing weight configurations.

Furthermore, since model performance varies across classes, I extended this approach to a class-wise ensemble, where weights are learned independently for each class. This enables the ensemble to exploit model specialization, assigning greater importance to models that perform better on specific subsets of the data.

Finally, I introduced the specialist models trained on restricted subsets of classes corresponding to biologically meaningful groups (myeloid, lymphoid, and neutrophil lineages). These groups are particularly challenging due to fine-grained morphological differences between maturation stages. My intuition was that models trained specifically on these subsets could learn more discriminative features within each group than a global classifier.

At inference time, I integrated these specialists in a conditional refinement step. When the base ensemble predicted a class belonging to one of these groups, I queried the corresponding specialist. If the specialist’s confidence exceeded a learned threshold, I used its predictions to refine the probabilities within that group through a convex combination controlled by a weighting parameter. Both the blending coefficients and confidence thresholds were optimized on the validation set.

7.3 Greedy Class-wise Ensembling vs. Dirichlet Weight Search

In addition to Dirichlet Weight Search, I implemented a class-wise greedy ensemble that iteratively refines predictions by selectively blending outputs from generalist and specialist models. At each

step, candidate updates are evaluated—either incorporating predictions from another general model for a specific class or integrating specialist predictions over predefined class groups—and the update that maximizes macro F1 on the validation set is applied. This process is repeated until no further improvement is observed.

Interestingly, while the greedy class-wise ensembling strategy exhibited clear overfitting, the Dirichlet-based approach did not suffer from this issue to a significant extent, despite operating over a similarly rich and expressive search space. This indicates that the difference in performance is primarily driven by the optimization dynamics rather than the representational capacity of the ensemble.

The greedy method performs sequential, locally optimal updates, repeatedly using the same validation set to refine the ensemble. As a result, it effectively behaves as a high-capacity learner trained on the validation data, progressively fitting not only signal but also noise. The accumulation of many small improvements leads to a form of path-dependent overfitting, where early decisions influence subsequent updates and cannot be easily corrected.

In contrast, the Dirichlet search samples global weight configurations and evaluates them independently, without iterative refinement. This makes it closer to a model selection procedure rather than a training process, significantly reducing the risk of overfitting. Since each candidate solution is evaluated only once, the method is less prone to exploiting fine-grained noise patterns in the validation set.

Overall, this comparison highlights that, even when the search space is equally expressive, the choice of optimization strategy plays a crucial role in determining generalization performance in ensemble construction.

7.4 Analysis of Individual Models and Ensembles

Individual models. The standalone models achieve macro F1-scores in the range of 0.66–0.72, with the best performance obtained by the Swin-based model (0.715) and the strongest ConvNeXt configuration (0.706). Despite similar overall scores, their per-class behavior differs substantially.

Across all models, performance is near-perfect on dominant and morphologically distinct classes such as BA, EO, LY, and SNE. In contrast, the main difficulties arise in:

- **Neutrophil subtypes (BNE vs SNE)**, where recall for BNE is consistently low.
- **Myeloid lineage (MMY, MY, PMY)**, which exhibits strong overlap due to continuous maturation.
- **Rare classes (PLY, PC)**, where instability is driven by extremely small sample sizes.

Importantly, different models exhibit different strengths. For instance, some models achieve higher recall on BNE (e.g., Model 7 in notebook), while others better separate myeloid stages (e.g., Model 2 in notebook). This confirms that the models are complementary rather than redundant as mentioned earlier.

Effect of ensembling. A simple weighted ensemble already yields a substantial improvement, increasing macro F1 from ~ 0.71 (best single model) to 0.765. This gain is consistent across most difficult classes, particularly:

- Improved balance between precision and recall for BNE.
- More stable predictions for myeloid subclasses.
- Slight gains on minority classes such as VLY.

The class-wise ensemble further improves performance to 0.772, confirming that different models are indeed better suited for different classes. Allowing class-specific weighting enables the ensemble to exploit this specialization more effectively.

Specialist models. Introducing specialist models yields only a modest improvement on the validation set (0.774), but leads to a consistent gain on the test set. This suggests that specialists primarily contribute to robustness rather than raw validation performance.

Greedy class-wise ensemble. The greedy class-wise optimization achieves the best validation performance (0.779). However, this improvement does not translate to the test set, where performance degrades compared to simpler ensembles.

This behavior indicates that the greedy procedure **overfits to the validation set**. By iteratively optimizing small class-wise adjustments, it captures noise and idiosyncrasies specific to the validation distribution, resulting in reduced generalization.

Generalization and final performance. In contrast, both the simple weighted ensemble and the class-wise ensemble generalize well, achieving similar performance on validation and test data. Notably, incorporating specialists—while only marginally beneficial on validation—consistently improves test performance, leading to a final macro F1-score of approximately **0.78**.

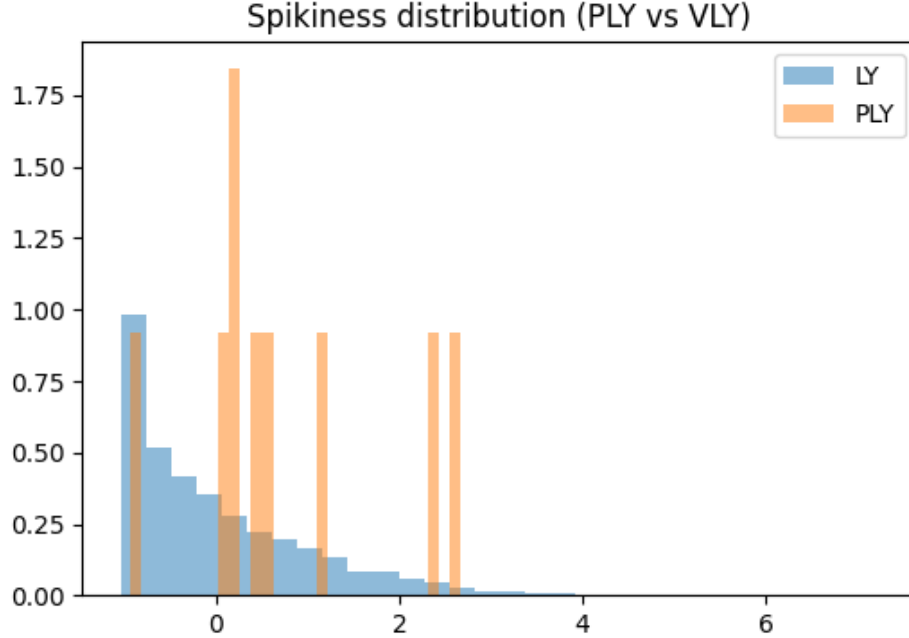
Class	Precision	Recall	F1-score	Support
BA	0.984	0.968	0.976	62
BL	0.897	0.950	0.923	302
BNE	0.420	0.627	0.503	59
EO	0.969	0.977	0.973	129
LY	0.964	0.956	0.960	1215
MMY	0.732	0.556	0.632	54
MO	0.903	0.927	0.915	412
MY	0.662	0.803	0.726	66
PC	0.833	0.500	0.625	10
PLY	1.000	0.500	0.667	2
PMY	0.667	0.471	0.552	17
SNE	0.990	0.973	0.981	1953
VLY	0.647	0.600	0.623	55
Macro Avg	0.822	0.754	0.774	4336
Weighted Avg	0.946	0.942	0.943	4336

Table 2: Classification report for the final ensemble with specialists on the validation set.

8 Future work: Biological Heuristics

Inspired by [5], I investigated whether biologically-inspired morphological features could meaningfully separate the most problematic classes, particularly PLY and LY. To this end, I analyzed the distribution of a spikiness score derived from contour irregularity. While a between the class distributions is observable, the overlap remains substantial, indicating limited discriminative power in my setting.

Importantly, this should not be interpreted as evidence that spikiness is inherently uninformative. On the contrary, [5] demonstrate that geometric spikiness can significantly improve performance when properly integrated. Instead, the discrepancy is likely due to the quality of the underlying morphological estimation. While their approach operates on CellPose (which we were not allowed to use as it’s pretrained on cell datsets), I rely on a fully handcrafted pipeline based



on classical computer vision techniques, including K-means-based nucleus estimation and contour-based geometry. Although interpretable, this approach is inherently limited in capturing subtle morphological and textural cues.

Overall, these results suggest that the effectiveness of spikiness as a discriminative feature depends critically on the accuracy of the underlying segmentation and the context in which it is applied. Improving cell and nucleus delineation, or integrating these features within a conditional refinement framework, represents a promising direction for future work.

9 Conclusion

This work addressed the challenging problem of fine-grained white blood cell classification under strong class imbalance and high inter-class similarity. Rather than relying on a single modeling approach, the methodology followed a structured and iterative process, where each design choice was guided by both empirical observations and an understanding of the underlying biological and statistical properties of the data.

At the representation level, deep convolutional and transformer-based models (ResNet-50, ConvNeXt, and Swin Transformer) proved highly effective feature extractors, consistently achieving strong performance. However, analysis of confusion matrices revealed that errors were not uniformly distributed, but concentrated within biologically related groups, reflecting intrinsic ambiguity in the task rather than purely architectural limitations.

This observation motivated a shift in focus from improving individual models to exploiting their complementarity. Although different models achieved similar overall performance, they exhibited distinct error patterns, particularly on difficult classes such as neutrophil subtypes and myeloid maturation stages. Leveraging this diversity through ensembling led to substantial gains.

A simple weighted ensemble already produced a significant improvement over individual models, confirming that diversity in architectures and preprocessing pipelines translates into complementary predictive behavior. Extending this approach to class-wise weighting further improved performance,

demonstrating that different models are better suited to different regions of the label space.

The introduction of specialist models provided an additional layer of refinement. While their impact on validation performance was modest, they consistently improved test performance, indicating increased robustness and better handling of fine-grained distinctions. This highlights an important point: improvements that are not immediately visible on validation data can still contribute to generalization when they address structurally difficult aspects of the problem.

In contrast, more complex optimization strategies such as greedy class-wise ensembling achieved higher validation scores but failed to generalize, illustrating the risk of overfitting when repeatedly optimizing on the same validation set. This emphasizes that the effectiveness of an ensemble is not only determined by its expressive power, but also by the stability of the optimization procedure used to construct it.

The final system, combining diverse architectures, carefully designed preprocessing, and a well-regularized ensemble strategy with specialist refinement, achieved a macro F1-score of approximately 0.78 on the test set. This result reflects a balance between performance and robustness, rather than aggressive optimization.

Overall, this work demonstrates that in complex, fine-grained classification tasks, improvements are less about finding a single superior model and more about understanding the structure of errors and combining complementary sources of information. Future work could further explore stronger integration of biological priors, improved morphological feature extraction, and more principled methods for incorporating specialist knowledge into the ensemble.

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